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Evaluating Model Sensitivity of Various Data Input Resolutions for Large Scale Flood Inundation Modeling for Hurricane Helene, NC

Joshua Smallwood  | Robert Pennell  | Emily S. Ondich 

US Army Engineer Research and Development Center, Coastal and Hydraulics Lab, Vicksburg, Mississippi, USA

Correspondence: Joshua Smallwood (joshua.j.smallwood@usace.army.mil)

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ABSTRACT

This study applies a statistically sound methodology for contrasting real-valued datasets against simulated results generated from the US Army Engineer Research and Development Center's Coastal and Hydraulics Laboratory's first-order flood inundation hydrologic modeling suite, AutoRoute, to determine the impact of spatial resolution on flood inundation extent, velocity, depth, and top-width estimates. Nine AutoRoute simulations were conducted for Buncombe County, NC, for flooding events resulting from Hurricane Helene in September 2024. Each AutoRoute simulation was run with a unique combination of spatial resolutions for two of its primary input products: digital elevation model (DEM) and land cover (LC). The velocity, depth, and top-width (VDT) outputs were tested for distributional nonoverlap by calculating Cliff's δ estimate, the Spatial Bhattacharyya Coefficient (SBC), and polygonal overlap with NASA-JPL's Observational Products for End-Users from Remote Sensing Analysis observed flood extents. The results confirm DEM products more heavily influence VDT output than LC, and lower input product resolution causes lower output velocity values and higher output top-width values, potentially resulting in velocity underestimation and top-width overestimation. This methodology offers an efficient and systematic evaluation of real-valued outputs and can be applied to additional areas of interest, flood extents, flow scenarios, and input products of varying spatial resolutions.

1 | Introduction

As the availability of new global terrain products continues to grow, the US Army Engineer Research and Development Center's Coastal and Hydraulics Laboratory (ERDC-CHL) has interest in improving and quantifying the impacts that input product spatial resolution have on first-order flood inundation models. Quantifying these impacts will allow modelers to make informed decisions on which available data to utilize. Developed by ERDC-CHL, AutoRoute is a first-order regional to continental scale flood mapping tool that offers improved computational efficiency and provides reasonable first-order approximations of stream characteristics and flood inundation extents by solving Manning's equation. AutoRoute samples

cross sections along perennial and ephemeral streams and rivers to assess flood inundation for general research collaboration and decision support capabilities for government organizations. 10–30 m resolution digital elevation model (DEM) data are the most commonly used input products, given the model's global use, where high-resolution elevation products may not be uniformly available. Users have historically used 30-m Land Cover (LC) for applications in the contiguous United States (CONUS) and 1-km LC for applications outside of the contiguous United States (OCONUS). More recently, 10 m landcover products based on Sentinel-1 and 2 data have become available with global coverage (Karra et al. 2021; Zanaga et al. 2022). The highest resolution DEM and LC available for a given area of interest (AOI) are generally preferred.

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AutoRoute is highly influenced by its input products, and there is little quantified risk on the effects of input products' spatial resolution on model accuracy (Follum et al. 2020). Identifying AutoRoute's sensitivity to changes in its input products' spatial resolutions can be helpful in determining model accuracy for AOIs with only lower resolution data available. Additionally, due to the computational burden required for higher resolution AutoRoute simulations, quantifying model sensitivity may allow for lower resolution AutoRoute simulations with a lighter computational burden without sacrificing model accuracy.

In 2019, AutoRoute simulations using higher resolution 3-m DEM were compared to initial simulations using 9-m DEM for test locations in Mississippi and North Carolina. In this initial comparison, Follum contrasted F values to conclude that higher resolution 3-m DEM did not substantially improve flood inundation results from lower resolution 9-m DEM (2020). The area for each flood extent output was also calculated to compute percent error when compared to USGS observed data. Our study expands upon Follum's work and considers how changes in two of AutoRoute's primary input data layers, DEM and LC, affect primary outputs velocity (V), depth (D), top-width (T), and flood extent. We also introduce a joint SBC and Cliff's δ framework that separates uncertainty driven spatial differences from input product resolution

effects. The objective of this study is to answer the following primary research questions:

1. How is AutoRoute sensitive to changes in its input products' spatial resolutions?
2. At which resolutions of DEM and LC input products is AutoRoute most accurate when compared to observed flooding data?

2 | Methodology

2.1 | Study Area

To quantify AutoRoute's sensitivity to changes in its input products spatial resolutions, the area around Asheville, North Carolina was selected as the study domain due to the recent flooding as a result of Hurricane Helene. Hurricane Helene hit western North Carolina on September 27, 2024, bringing historic rainfall and flooding to the area. Over 500 mm of rainfall fell on the region, categorizing the event as greater than a 1000-year storm, meaning it has a less than 0.1% chance of occurring each year. Rivers surrounding Asheville and throughout western North Carolina hit record-breaking peak flows (Amorim et al. 2025). The AOI analyzed in this study spans Asheville and the surrounding Buncombe County, as shown in Figure 1 below.

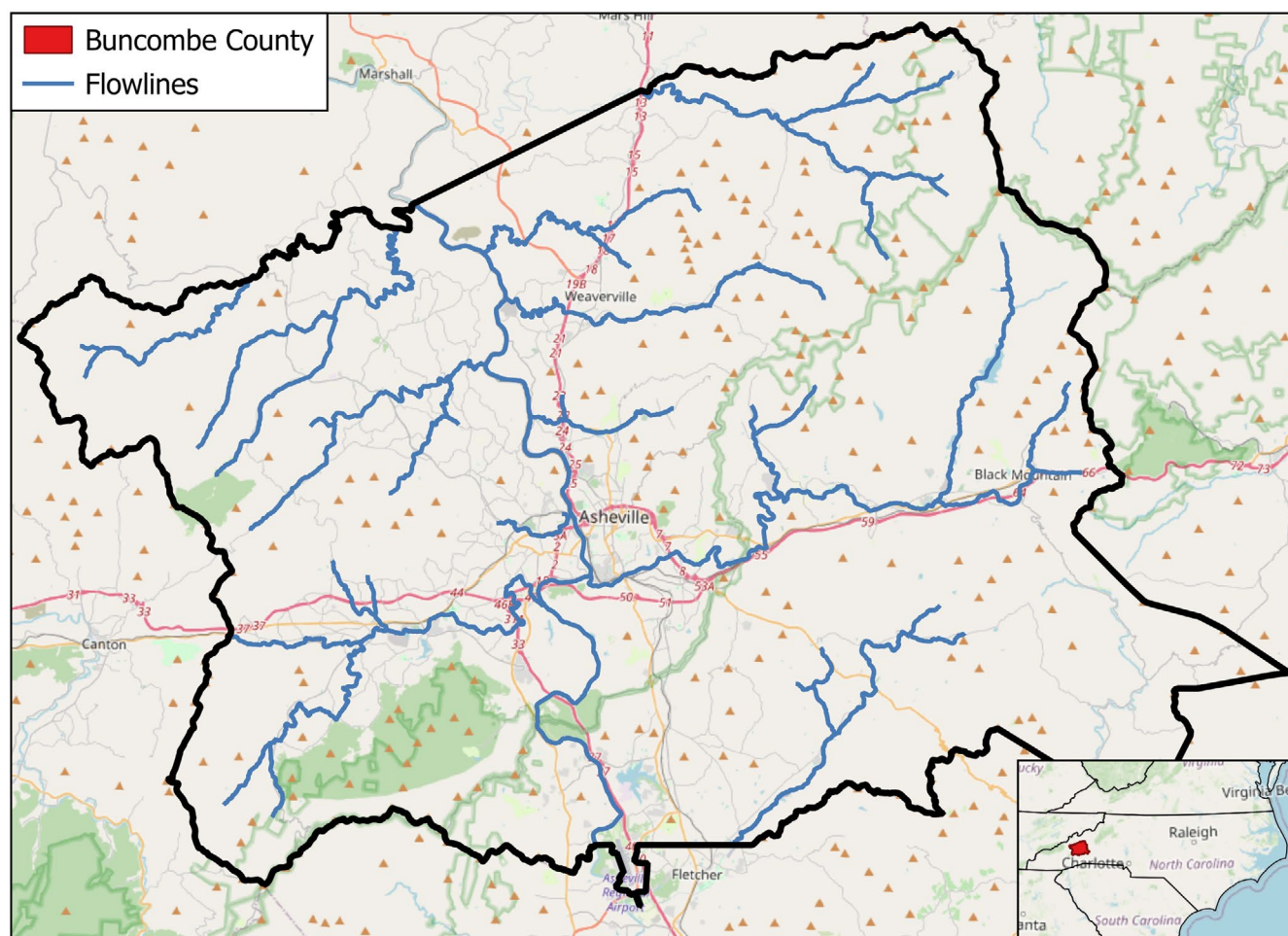


FIGURE 1 | Flow map of Buncombe county, NC.

2.2 | Models and Data

Ingesting DEM, land cover, stream network, and stream flow data, AutoRoute creates flood inundation maps by sampling cross sections for each stream cell of an input DEM, solving Manning's equation to determine the water's average velocity at that cross section, calculating a normal depth at any given area, and determining which cross sections are projected to experience flooding. Only the cells aligning with cross sections of the map that are projected to flood are filled in on the raster format. FloodSpreader, a complementary code for visualization of the flood extent, is then applied to the raster to fill in gaps surrounded by flooding cells, omit outlier cells that are surrounded by non-flooding cells, and smooth the borders of the floodplain (Follum et al. 2017).

Retrospective daily streamflow data obtained from GEOGLOWS for 29 September 2024 were used in this study (GEOGLOWS 2025). While peak flows were observed in the study area on 27 September, peak flows didn't appear in the GEOGLOWS streamflow time series dataset until 2 days later. The retrospective streamflow product is bias corrected with the Stream Analysis for Bias Estimation and Reduction (SABER) method (Hales et al. 2023). DEM and LC data spanning the AOI were collected from the sources listed in Table 1.

Each DEM considered in this study was generated using different source data and post processing techniques. The source DEM from the North Carolina Spatial Data Download is a LIDAR derived dataset, with an initial resolution of 3.125 ft which we resampled to meters, resulting in 0.95-m resolution, subsequently labeled as 1-m. All vegetation and building points are filtered out, and the remaining ground classification points are interpolated to create a continuous ground level surface (North Carolina Emergency Management 2025). This dataset has a mean non-vegetated vertical accuracy of +0.027 m and a mean vegetated vertical accuracy of +0.050 m (North Carolina Emergency Management 2018). The 12-m TanDEM-X dataset is derived from synthetic aperture radar collected by the German Aerospace Centre and Airbus Defense and Space. A Landsat-derived tree canopy cover product was used to remove vegetation bias from the DEM. Following vegetation bias removal, a smoothing algorithm was used to remove any remaining artifacts as the algorithm holds peaks and valleys in place to ensure the dataset is hydrologically accurate. The TanDEM-X dataset has a vertical accuracy of ($\pm$) 0.2 m (Wessel et al. 2018). The 30-m FABDEM is a

version of the Copernicus GLO-30 DEM in which forests and buildings were removed to provide a bare earth DEM (Hawker et al. 2022). After forest and building removal, smoothing was also completed, resulting in a DEM with a mean vertical accuracy of +0.33 m. In comparison of accuracy with other publicly available 30-m global DEMs, FABDEM performs the best (Meadows et al. 2024). The landcover datasets utilized in the study include: the European Space Agency (ESA) WorldCover which provides global 10-m resolution landcover data collected by Sentinel-1 and Sentinel-2, the National Land Cover Dataset (NLCD) which is an annually produced 30-m resolution land cover dataset spanning the continental US, and the Advanced Very High Resolution Radiometer (AVHRR) Global Land 1-km dataset which spans the continental US and is produced by USGS based on data collected by the TIROS satellite program.

The varying combinations of DEM and LC were coupled with the GEOGLOWS flows for every AutoRoute simulation, generating velocity, depth, and top-width values (VDT) for each stream segment throughout the AOI. Flood extent polygons were then produced by FloodSpreader, intaking the VDT output from AutoRoute and converting it into a polygon for visualization. After obtaining velocity, depth, and top-width outputs for each of the nine AutoRoute simulations, the data were sorted and labeled with a numerical Run ID from 1 to 9. Table 2 shows the experiment design matrix, identifying each

TABLE 2 | Run ID and input product spatial resolution.

Run ID	DEM resolution (in meters)	LC resolution (in meters)
1	1	10
2	1	30
3	1	1000
4	12	10
5	12	30
6	12	1000
7	30	10
8	30	30
9	30	1000

TABLE 1 | DEM and LC product source locations.

DEM resolution (m)	Source	LC resolution (m)	Source
1	North Carolina Spatial Data Download ^a	10	ESA WorldCover ^d
12	TanDEM-X Noise Reduced ^b	30	USGS NLCD ^e
30	FABDEM ^c	1000	AVHRR ^f

^aNorth Carolina Emergency Management, 2017.

^bWessel, et al. 2018

^cNeal, J. and Hawker, L., 2023.

^dEuropean Space Agency WorldCover, 2021.

^eUSGS, 2024.

^fUSGS, 2018.

AutoRoute Run ID by the spatial resolutions of its DEM and LC input products.

All data, with the exception of the TanDEM-X Noise Reduced DEM provided by the National Geospatial-Intelligence Agency (NGA), is publicly available online.

2.3 | Effect Size Measures

The two primary statistical tests used in this study are Cliff's δ estimate and the Spatial Bhattacharyya Coefficient. Both are measures of effect size that compare data occurrences between two independent groups. Cliff's δ calculates the measure of distributional nonoverlap between any two given distributions. The test statistic ranges from -1 to 1 , with the more extreme values indicating significant differences in data occurrences and values closer to 0 indicating closer occurrences in the two datasets (Meissel and Yao 2024). Cliff's δ is a robust, non-parametric measure of effect size, not requiring assumptions of normality and homoscedasticity. However, geospatial data is highly spatially autocorrelated, so this study utilizes Cliff's δ to only determine the magnitude and direction of nonoverlap between datasets (Fiel 2026).

$$\text{Cliff's delta } (\delta) = \frac{1}{n_x n_y} \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} \text{sign}(x_i - y_j)$$

Complementary to the Cliff's δ , the Spatial Bhattacharyya Coefficient (SBC) provides a distribution-based measure of spatial similarity that is explicitly designed to account for geographic structure in the data. The SBC evaluates distributional similarity between neighboring or overlapping spatial units rather than all possible pairs. This is accomplished through the use of a spatial weights matrix, which restricts comparisons to locations that are geographically proximate, preserving the assumption of spatial dependence inherent in geospatial datasets. As a result, SBC quantifies the extent to which nearby spatial units contain similar distributional characteristics, producing a single coefficient based on the probability of overlap rather than rank-based dominance. While Cliff's δ emphasizes the direction and magnitude of differences between groups, SBC captures how closely the spatial distributions of the two datasets align. Both measures directly address the primary research questions by quantifying the overlap between each pair of datasets and the magnitude and direction of the differences.

$$\text{SBC} = \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} BC(i, j)}{\sum_{i=1}^n \sum_{j=1}^n w_{ij}}$$

where w_{ij} is an element of a spatial weights matrix that has a binary weight, n is equal to the total number of observations, and the sum of the spatial weights matrix over all observations with itself and its neighbors is equal to n (Koo et al. 2019). The SBC ranges from 0 to 1 , with a higher value indicating closer overlap between the two distributions. The SBC is specifically relevant to this study because it explicitly accounts for error and spatial autocorrelation.

Next, Cliff's δ and SBC estimates were calculated for each of the comparisons between the 9 independent datasets that

had overlapping cells for each of V, D, and T. Accuracy of the AutoRoute produced flood extents was analyzed by quantifying the percent overlap of each modeled flood extent with observed flood extent from the September 2024 flooding. Observed data, collected on 30 September to 3 October, was provided by the National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory's (JPL) Observational Products for End-Users from Remote Sensing Analysis (OPERA). The OPERA flood extent dataset was built from radar data collected by Sentinel-1 (NASA-JPL 2024).

3 | Results

3.1 | Effect Size Results

Cliff's δ does not have inherent absolute thresholds in spatially autocorrelated environmental data, so its values in this study are interpreted relatively rather than categorically. Thresholds commonly cited in the behavioral sciences are based on independent, bounded, and approximately normal data, which may not translate into geospatial data (Getis 2008, 55). Here, Cliff's δ is used to primarily assess the direction of input resolution dominance between datasets, rather than to label differences as small, medium, or large according to external benchmarks. Table 3 contains Cliff's δ and SBC with 95% confidence intervals truncated at four decimal places for each of the AutoRoute simulation comparisons for V, D, and T. It also contains the number of overlapping cells from the two distributions, with a higher N_{overlap} indicating a similar data makeup of Row and Column IDs. A color gradient was added to the two test statistics, with Cliff's δ having a red color fill for values closer to the extremes, -1 and 1 indicating a higher number of sign differences, and green at its center 0 indicating fewer. The SBCs have a red color fill for values closer to 0 and green closer to 1 .

In addition to the calculation of Cliff's δ and SBC as measures of effect size, the raw VDT outputs were plotted to visualize the differences in VDT outputs for each Run ID. The violin plots for each are shown in Figure 2 below.

3.2 | Accuracy Results

The table below displays the percentage that each run's output flood extents overlap with OPERA observed flood extents. Run 1 (1-m DEM and 10-m LC) saw the most overlap while Run 9 (30-m DEM and 1000-m LC) saw the least. Overall, an increase in resolution caused the model to better align with the observational data. Figure 3 below displays the coverage of both OPERA data and modeled flood extents across the study area. Given the overall percent overlap of the initial flood extent results, a subset of the study area with the highest coverage of OPERA data was run in an effort to provide better context to the initial results.

The subset analysis provided much higher overlap than the initial results. In this segment of the French Broad River, Run 7 (30-m DEM and 10-m LC) aligned the best while Run 6 (12-m DEM and 1000-m LC) had the least amount of overlap. While

TABLE 3 | Cliff's δ and SBC for velocity, depth, top-width.

Run 1	Run 2	Variable	N_{overlap}	Cliffs δ	SBC	SBC CI
VDT1	VDT2	V	339,829	0.0883	0.9999	(0.99995, 0.99998)
		T	339,829	-0.0192	0.9995	(0.9989, 0.9995)
		D	339,829	-0.0358	0.9999	(0.99996, 0.99998)
VDT1	VDT3	V	332,018	0.4983	0.9999	(0.99994, 0.99997)
		T	332,018	-0.1360	0.9951	(0.99497, 0.9955)
		D	332,018	-0.1969	0.9999	(0.99995, 0.99998)
VDT1	VDT4	V	98	0.5118	0.7717	(0.5905, 0.7752)
		T	98	-0.4604	0.9110	(0.7115, 0.9272)
		D	98	-0.1688	0.8724	(0.7603, 0.8907)
VDT1	VDT5	V	99	0.5073	0.7185	(0.5258, 0.7395)
		T	99	-0.4676	0.9015	(0.7307, 0.9231)
		D	99	-0.1712	0.8385	(0.7278, 0.8688)
VDT1	VDT6	V	92	0.7648	0.5374	(0.3237, 0.5602)
		T	92	-0.5207	0.8412	(0.6489, 0.8781)
		D	92	-0.3304	0.8100	(0.6501, 0.8418)
VDT1	VDT7	V	37	0.7852	0.3876	(0.1061, 0.4853)
		T	37	-0.2330	0.7058	(0.2619, 0.7424)
		D	37	0.5675	0.6843	(0.3794, 0.7485)
VDT1	VDT8	V	36	0.7469	0.3187	(0.0758, 0.3750)
		T	36	-0.1496	0.7655	(0.1747, 0.7854)
		D	36	0.5848	0.6477	(0.3714, 0.7403)
VDT1	VDT9	V	40	0.9137	0.2957	(0.0580, 0.3611)
		T	40	-0.3800	0.6717	(0.2471, 0.6999)
		D	40	0.3750	0.6492	(0.4656, 0.7274)
VDT2	VDT3	V	342,052	0.3134	0.9999	(0.99994, 0.99997)
		T	342,052	-0.1015	0.9957	(0.9956, 0.9959)
		D	342,052	-0.1434	0.9999	(0.99994, 0.99997)
VDT2	VDT4	V	104	0.3544	0.8446	(0.6583, 0.8628)
		T	104	-0.4467	0.9129	(0.7305, 0.9310)
		D	104	-0.1614	0.8937	(0.8132, 0.9140)
VDT2	VDT5	V	106	0.3837	0.8705	(0.6586, 0.8811)
		T	106	-0.4546	0.9099	(0.7696, 0.9318)
		D	106	-0.1965	0.8852	(0.7745, 0.9038)
VDT2	VDT6	V	101	0.6276	0.7058	(0.5239, 0.7554)
		T	101	-0.5259	0.8408	(0.6694, 0.8759)
		D	101	-0.3537	0.8499	(0.6693, 0.8798)

(Continues)

TABLE 3 | (Continued)

Run 1	Run 2	Variable	N_{overlap}	Cliffs δ	SBC	SBC CI
VDT2	VDT7	V	39	0.8211	0.4392	(0.1504, 0.5205)
		T	39	−0.2583	0.6356	(0.2312, 0.6496)
		D	39	0.5838	0.6978	(0.4271, 0.7676)
VDT2	VDT8	V	37	0.7086	0.3327	(0.1044, 0.4366)
		T	37	−0.1088	0.7182	(0.1912, 0.7547)
		D	37	0.6157	0.6109	(0.3512, 0.7089)
VDT2	VDT9	V	41	0.9381	0.2474	(0, 0.3304)
		T	41	−0.3277	0.5224	(0.1266, 0.5551)
		D	41	0.4116	0.7264	(0.5133, 0.8023)
VDT3	VDT4	V	104	0.0075	0.8741	(0.7022, 0.8622)
		T	104	−0.3143	0.9030	(0.7772, 0.9208)
		D	104	0.0723	0.8959	(0.7945, 0.9100)
VDT3	VDT5	V	105	0.0958	0.9008	(0.7543, 0.8955)
		T	105	−0.3353	0.9020	(0.7883, 0.9275)
		D	105	0.0017	0.8917	(0.7917, 0.9105)
VDT3	VDT6	V	99	0.4437	0.7433	(0.5960, 0.7703)
		T	99	−0.4121	0.8312	(0.7115, 0.8704)
		D	99	−0.1610	0.8980	(0.7688, 0.9095)
VDT3	VDT7	V	41	0.5859	0.5212	(0.2295, 0.6070)
		T	41	−0.1778	0.6684	(0.2039, 0.7162)
		D	41	0.7358	0.5919	(0.3387, 0.7087)
VDT3	VDT8	V	40	0.5787	0.4990	(0.2668, 0.5881)
		T	40	−0.0637	0.65369	(0.1738, 0.7352)
		D	40	0.7387	0.5489	(0.3360, 0.6652)
VDT3	VDT9	V	42	0.8174	0.5074	(0.1642, 0.5616)
		T	42	−0.3117	0.5112	(0.1652, 0.5638)
		D	42	0.5561	0.7277	(0.5147, 0.7838)
VDT4	VDT5	V	28,744	0.1103	0.9998	(0.9997, 0.9999)
		T	28,744	−0.0295	0.9978	(0.9970, 0.9979)
		D	28,744	−0.0308	0.9997	(0.9995, 0.9998)
VDT4	VDT6	V	28,601	0.5109	0.9987	(0.9983, 0.9989)
		T	28,601	−0.1776	0.9901	(0.9881, 0.9905)
		D	28,601	−0.2344	0.9982	(0.9977, 0.9985)
VDT5	VDT6	V	28,387	0.3887	0.9992	(0.9989, 0.9993)
		T	28,387	−0.1454	0.9926	(0.9913, 0.9928)
		D	28,387	−0.1796	0.9991	(0.9988, 0.9993)

(Continues)

TABLE 3 | (Continued)

Run 1	Run 2	Variable	N_{overlap}	Cliffs δ	SBC	SBC CI
VDT7	VDT8	V	13,749	0.0589	0.9999	(0.9933, 1)
		T	13,749	-0.0131	0.9992	(0.9978, 0.9990)
		D	13,749	-0.0320	0.9998	(0.9966, 0.9999)
VDT7	VDT9	V	13,607	0.4357	0.9998	(0.9995, 0.9999)
		T	13,607	-0.1369	0.9913	(0.9896, 0.9914)
		D	13,607	-0.2356	0.9972	(0.9967, 0.997)
VDT8	VDT9	V	13,491	0.3500	0.9999	(0.9997, 0.9999)
		T	13,491	-0.1217	0.9939	(0.9919, 0.9941)
		D	13,491	-0.1880	0.9982	(0.9975, 0.9986)

Note: Green indicated more overlap between runs, while red indicates less overlap between runs.

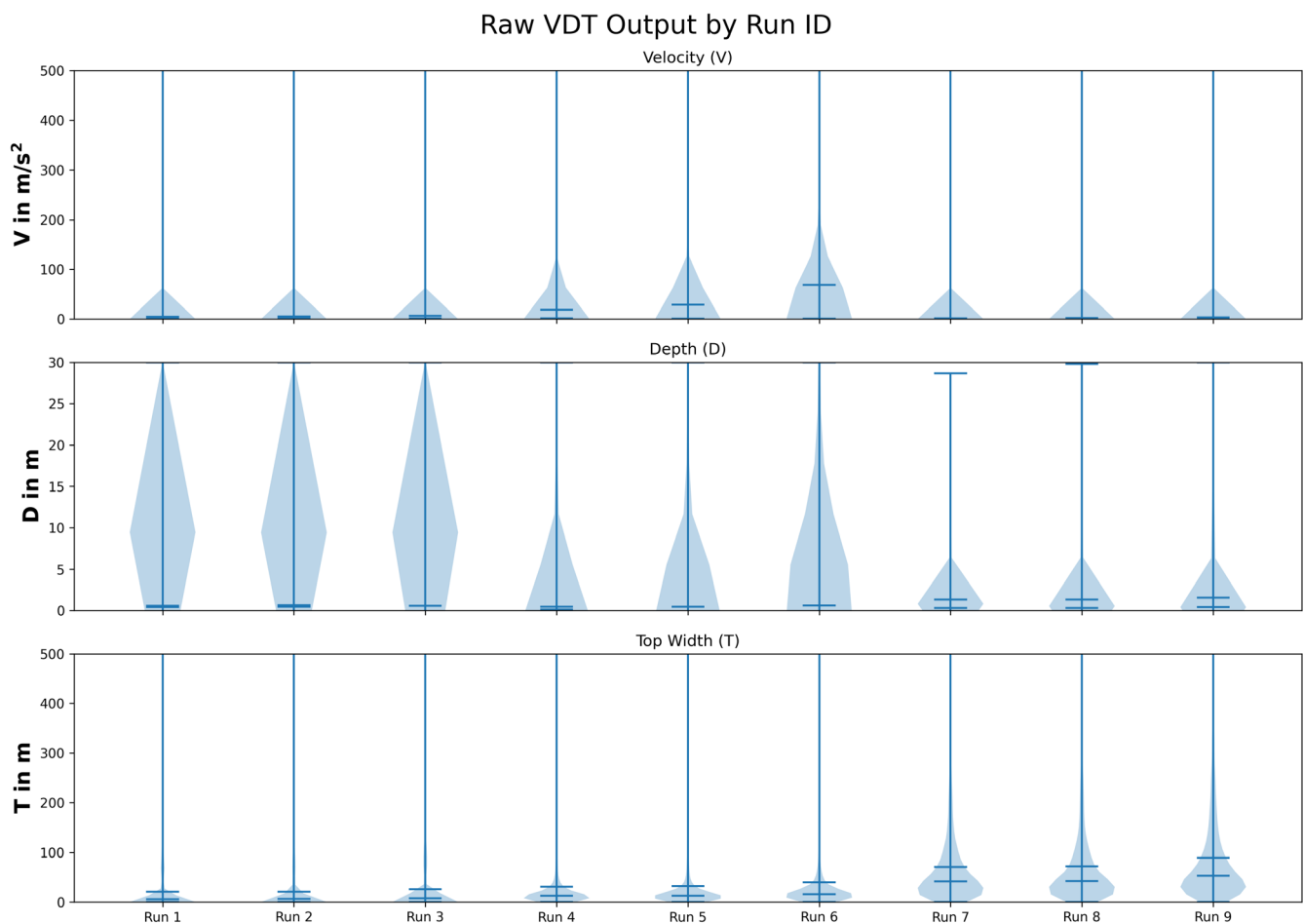


FIGURE 2 | Violin plot of raw VDT output by Run ID.

there appears to be no strong trend for DEM resolution in this subset analysis, there is a distinct trend among the LC resolution. Runs 3, 6, and 9 (1000-m LC) resulted in the lowest percentages for each of their respective DEM groups. While Runs 1, 4, and 7 (10-m LC) provided the best results for their respective DEM groups. This trend matches the full extent analysis, indicating that higher resolution LC performs better regardless of DEM. Despite the notable increase in accuracy, values likely remain

low due to a mismatch in data collection dates between OPERA flood extents (30 September to 3 October) and the GEOGLOWS flows (peak flows in the area occurred 27 September but did not generate in GEOGLOWS until 29 September) used in the modeling effort.

To explore the impact the temporal discrepancy between datasets was causing, AutoRoute was rerun using GEOGLOWS

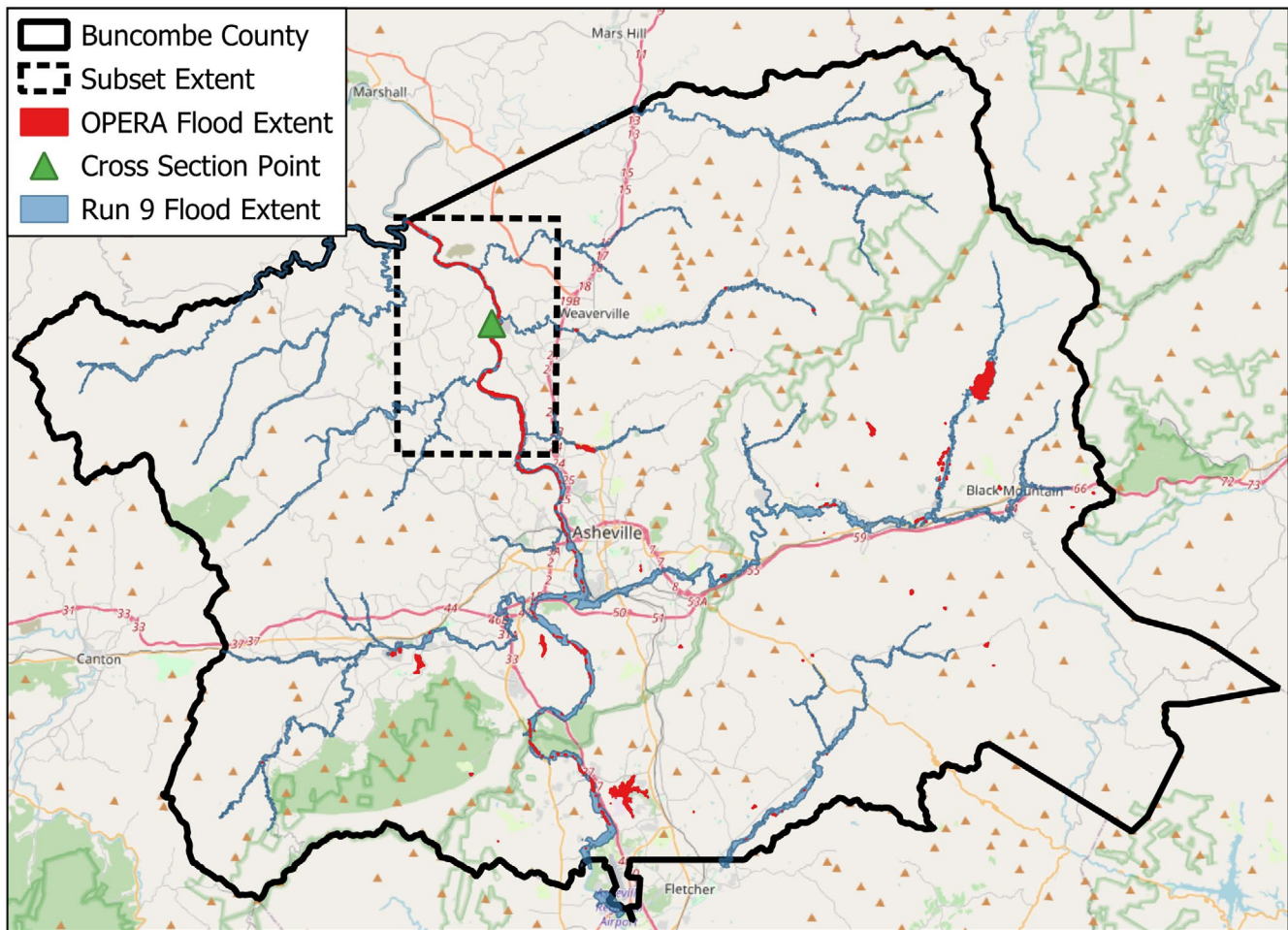


FIGURE 3 | Data extents for observed OPERA data and AutoRoute generated flood extents, and subset vs overall study area extents.

flows that matched OPERA collection dates. The average discharge from 2 October to 5 October was collected from GEOGLOWS, which matched the 30 September to 2 October OPERA data collection, as GEOGLOWS has a 2-day lag. The results from this analysis, shown in Table 4, improved percent overlap greatest in Runs 4–6 and 7–9, but only slightly improved Runs 1–3. Conclusions from the overall and subset flood extent analysis must be considered within the context of this tertiary flood extent analysis displaying the disparities due to different timing.

3.3 | Runtime Comparison

Table 5 below displays the runtime for each combination of input product resolutions. Runs 1, 2, and 3 which utilized the 1-m DEM produced significantly higher run times than runs utilizing the 12- and 30-m DEMs. LC resolution does not significantly alter run time as each grouping of DEMs varies minimally with different LC. While scaling up the modeling domain will increase runtime for all resolutions, the lower the resolution the lower the run time will be in comparison to higher resolution options. This result provides important context for understanding the tradeoffs between differing resolutions and computational requirements for large spatial domains.

TABLE 4 | Simulated and observed flood extent percent overlap.

Run ID	Percent overlap with OPERA data	Subset area percent overlap with OPERA data	Percent overlap using matching GEOGLOWS and OPERA collection dates
1	23.6844	52.5377	24.0421
2	22.3935	49.1552	24.5973
3	21.2958	49.0953	23.5001
4	14.6138	47.0154	23.1909
5	13.9457	40.4301	23.9049
6	12.5953	38.5895	24.0860
7	7.5078	54.9653	12.8067
8	7.3104	53.8849	13.3777
9	6.6929	45.7504	13.4920

4 | Discussion

While each of these methods of statistical analysis provides detailed quantitative evaluations of the raw VDT outputs generated

TABLE 5 | Runtime of each combination of resolutions.

Run ID	Runtime (min:sec)
1	14:14.00
2	14:04.00
3	15:21.00
4	00:22.38
5	00:21.91
6	00:20.78
7	00:12.08
8	00:12.61
9	00:12.69

by AutoRoute, the combined application of Cliff's δ , SBC, and flood extent percent overlap allows for better informed hydrologic inferences to be made. Primary differences among velocity, depth, and top-width output show systematic trends as DEM and LC input products' spatial resolution are varied.

4.1 | Differences Among V, D, and T

AutoRoute simulations output a single set of V, D, and T data for each run. While it is unlikely for users to run separate simulations with the intention of extracting just one of V, D, or T, there may be a parameter that may hold greater significance contextually in the user's analysis. Important conclusions can still be drawn regarding each output individually. Top-width (T) is the least affected parameter by changes in input products' spatial resolutions. Table 6 shows the sum of the absolute value of both Cliff's δ and the sum of the SBC for each of the three parameters.

The sum of the absolute value of top-width's Cliff's δ is the smallest, suggesting that top-width has the most overlap between each pair of datasets with differing input resolutions. Depth is closely behind while velocity varies significantly with the most nonoverlap between different runs.

4.2 | DEM and LC Effects

In addition to quantifying differences in input products' spatial resolution effects on V, D, and T separately, these tests inform on the differences between DEM and LC effects on VDT output together. DEM spatial resolution has a stronger influence on VDT output than LC. First, the violin plots of V, D, and T all show a natural grouping for runs 1, 2, and 3; 4, 5, and 6; and 7, 8, and 9. Every quantifiable measure computed in this study suggests that DEM resolution heavily influences AutoRoute output: larger magnitude Cliff's δ s and smaller SBCs occurred in run comparisons with different DEM inputs (Table 3), and AutoRoute percent overlap with OPERA data (Table 4) and simulation runtime (Table 5) both show a natural grouping primarily by DEM input resolution.

Although DEM more strongly influences AutoRoute output and flood extent accuracy, LC also appears to have meaningful effects.

TABLE 6 | Sum of absolute value of Cliff's δ and SBC by parameter.

	Cliff's δ	SBC
V	13.34822	19.51393
D	8.063787	22.71354
T	6.830217	22.73266

Cliff's δ estimates and the violin plots indicate that lowering the resolution of LC input tends to pick up the more extreme values for depth and top-width. In the D and T scatterplots, for example, the points seem to slightly shift upward when comparing runs in each DEM group: comparing run 3–2 and 1, run 6–5 and 4, and run 9–8 and 7. Additionally, most of the pairwise comparisons in which all measures of effect size show smaller magnitude differences are runs in the same DEM group with LC at 10-m and 30-m resolutions. This indicates that while LC effects are smaller than DEM effects, differences in using LC 10-m resolution versus LC 30-m resolution may be negligible, the jump from 30-m to 1-km land cover may result in significant and unwanted variation.

4.3 | Cliff's δ and SBC Interaction

In this study, Spatial Bhattacharyya Coefficient (SBC) and Cliff's δ are used together to distinguish between apparent spatial agreement caused by uncertainty and true hydrologic differences caused by input product spatial resolutions. Koo et al. (2019) conclude that high spatial autocorrelation can be caused by either true geospatial structure or by a large uncertainty. SBC incorporates uncertainty in its calculation but still cannot determine whether similarity between distributions is physically meaningful or just numerical overlap. Interpreting both test statistics together, however, one can further determine the source of distribution similarities. A relatively high SBC and low Cliff's δ indicate that neighboring cross sections have strongly overlapping VDT distributions, and neither run consistently produced larger or smaller values than the other. A low SBC and low Cliff's δ may indicate that neighboring locations appear dissimilar between runs, but neither resolution dominates the other. In this study, a low Cliff's δ and a high SBC signal the closest distribution overlap between runs. Again, this most strongly appears in runs within the same DEM group.

4.4 | Flood Extent

Figure 4 below displays satellite imagery from prior to (21 August 2023) and aerial imagery following (5 October 2024) the storm, approximately 1 week after peak flows occurred. Despite the delay from the date of the storm to the date of the image collection, there are noticeable flood impacts to the area immediately south of the Swannanoa River, a tributary of the French Broad River, where sediment and debris fill the streets and parking lots. The OPERA dataset captures surface water extent, and since flood waters had receded by its collection date, OPERA did not capture flood impacts at this location. Notably, Buncombe County has a number of other tributaries that experienced similar flooding not captured in the OPERA dataset but that were captured by the AutoRoute

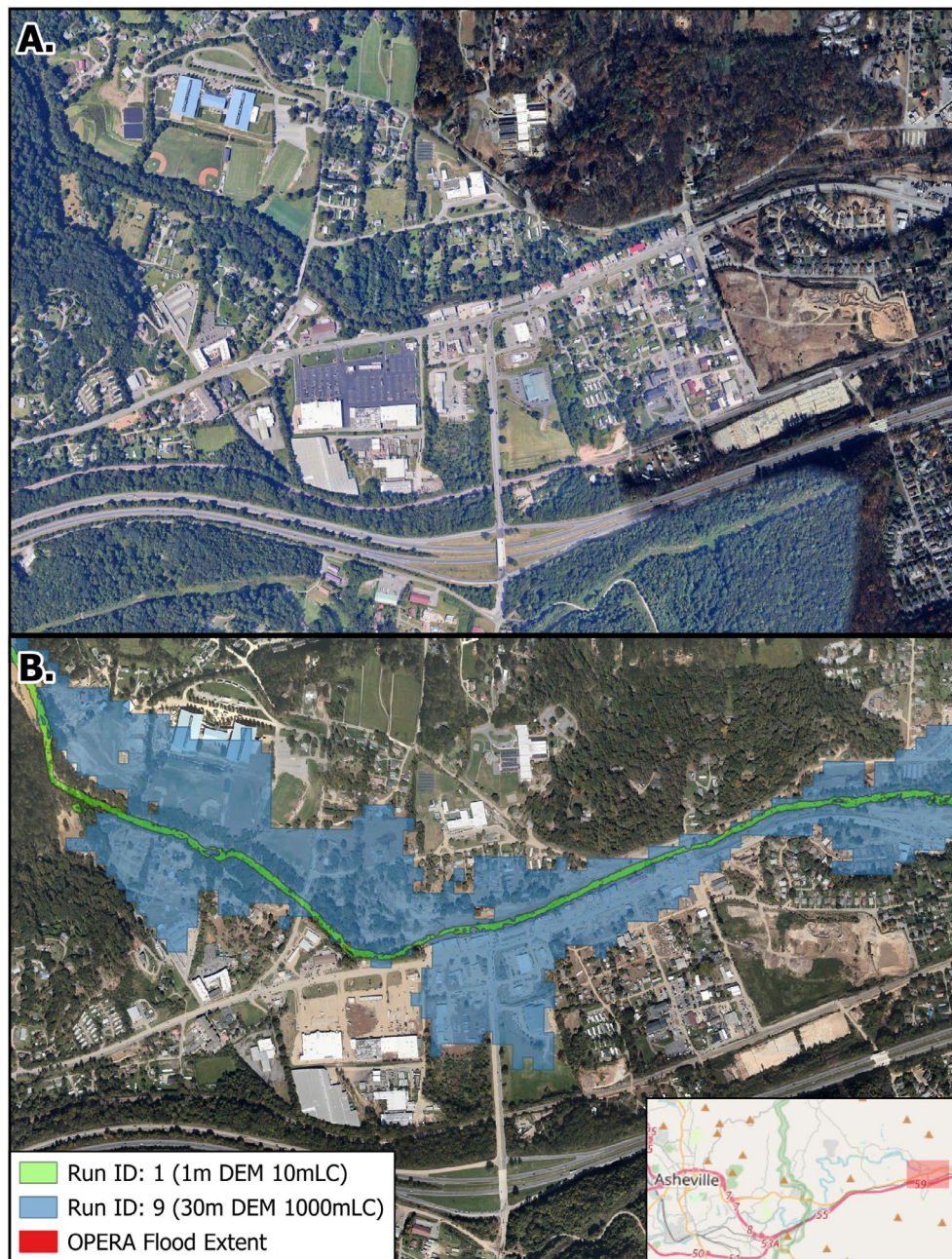


FIGURE 4 | Comparison of modeled and observed flood extents.

model. These discrepancies likely account for the relatively low percentage of overlap for both the full study area analysis and the matched timing analysis reported in Table 4. While the AutoRoute model produced out of bank results, flood extents are underestimated when compared to the extent of debris observed. This is likely due to a difference in magnitude between actual peak flows and GEOGLOWS simulated peak flows utilized in the model. USGS gage 03451000 just downstream of the area in Figure 4, (Swannanoa River at Bitmore, NC) estimated the peak flow at 60,800 cubic feet per second (cfs), while GEOGLOWS peak flow at the same location was only 4624 cfs (USGS 2025b).

This issue persists across the study area as USGS gage 03451500 (French Broad River at Asheville, NC) captured the peak of the storm at 111,000 cfs (USGS 2025a). In

comparison, GEOGLOWS peak flow at the same location was only 72,354 cfs. USGS stream gages calculate discharge from the measured gauge height, whereas GEOGLOWS models discharge based on the European Centre for Medium-Range Weather Forecasts Reanalysis 5 (ECMWF ERA5) climate data. These differing methods are probable causes of the discrepancy between input flow values. Despite this, GEOGLOWS provided the best available data for modeling peak flows given its fully encompassing spatial coverage of the AOI. Qualitative comparison between flood extents and imagery, as shown in Figure 4, suggests that higher resolution flood extents may underpredict observed flooding. This finding cannot be fully supported given the discrepancy in the magnitude of the simulated flows compared to reported streamflow from multiple USGS gauges in the area. The high resolution of the 1-m DEM picks up on smaller topographic features that are otherwise

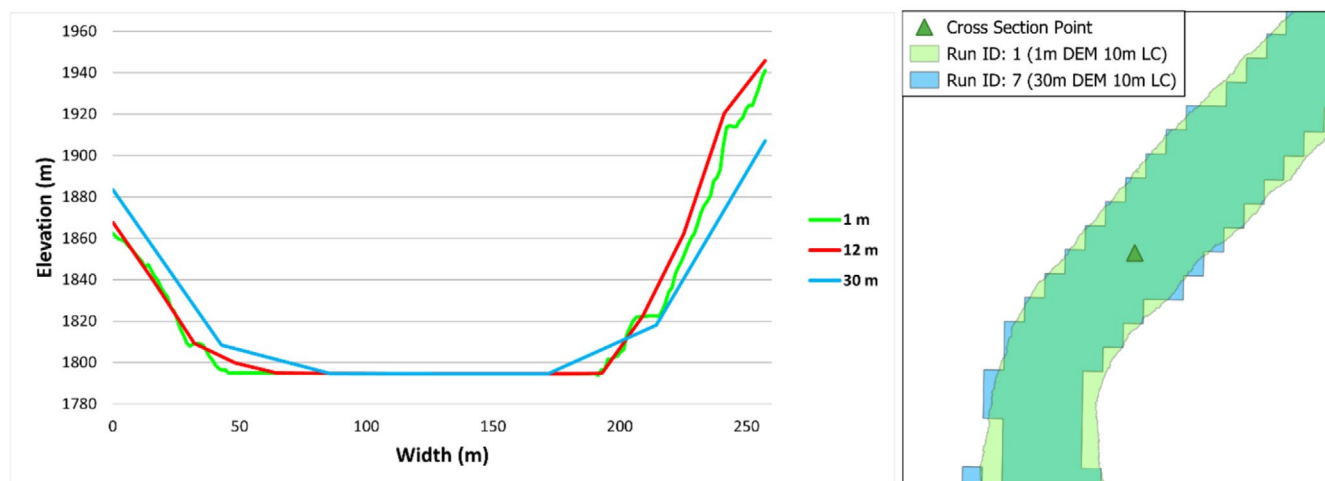


FIGURE 5 | Cross section and flood extent of French Broad River.

missed in the 30-m DEM (the difference in detail between DEMs is shown in Figure 5). The smoother profile and smaller bank slope of the 30-m DEM explains the larger flood extent observed in results using the 30-m DEM compared to the other products. An increase in flood extent would be expected from additional simulations using the 30-m DEM and streamflow more closely aligned with reported flows from the event, with a high potential to overpredict the final extent.

One possible cause for the limited trends among DEM resolutions in the subset analysis may be the lack of flooding in this reach of the French Broad River. Upon review of aerial imagery, less flooding impacts were seen in this reach, likely due to steeper forested banks which would have better contained higher flows. Figure 5 below displays the cross section at a point along this subset analysis area identified in Figure 3. The 30-m DEM provides the narrowest cross section while the 12-m and 1-m provide similar widths. Furthermore, the lower resolution of the 30-m DEM provides a coarser flood extent marked by jagged edges as seen in Figure 5. The 1-m DEM results in a smoother flood extent that fills in the jagged edges of the 30-m extent, producing a broader result.

Another potential contributor to the discrepancy in the results is the limitations of the AutoRoute model. To run efficiently over large areas, AutoRoute utilizes simplified inputs (DEMs, LC, and streamflow). Higher fidelity inland flooding and hurricane models incorporate additional parameters such as wind, soil moisture, storm track, size, and timing (Yang et al. 2019). Notable infrastructure impacts, including dams and reservoirs, are not inherently included in AutoRoute or many hurricane models but may contribute to further discrepancies in output. While GEOGLOWS does not directly consider infrastructure impacts either, its bias corrections aim to account for them (Hales et al. 2023).

Given that the comparison of modeled flood extents to OPERA flood extents proved to be less conclusive than anticipated, analysis of the relative percent change in area between each combination of runs was completed and summarized in Table 7. This analysis compared the percent change from higher resolution runs to lower resolution runs to quantify the changes input

resolution has on output flood extent. While these values may vary at other AOIs, they provide insight into the potential variation between using higher or lower resolution datasets when data availability becomes more restricted. The highest percent change (278.05%) was between runs 1 and 9—the runs with the highest (1-m DEM and 10-m LC) and lowest (30-m DEM and 1-km LC) input product resolutions. The smallest percent change (3.11%) was between runs 7 (30-m DEM and 10-m LC) and 8 (30-m DEM and 30-m LC).

When comparing between runs 1, 2, 3 (1-m DEM), 4, 5, 6 (12-m DEM), and 7, 8, 9 (30-m DEM), percent change remains below 13.45% with the exception of runs 4 and 6 which have a 27.12% change. This indicates that LC resolution does not have as much of an impact on flood extent output as DEM resolution, as runs with the same DEM resolution but different LC have minimal variation, whereas percent change increases drastically when comparing between varying DEM resolutions. These findings are consistent with the analysis of VDT outputs.

5 | Conclusion

Reasonable and informed recommendations can be made to maximize AutoRoute VDT output accuracy without the computational burden of necessarily having to use the highest resolution data available for every scenario. The combination of the effect size measures and percent-overlap comparisons between AutoRoute flood extents and OPERA flood extents confirms the assumption that the VDT output may vary substantially based on input product spatial resolution. More specifically, the largest measures of VDT nonoverlap and dataset dominance occurred between Run 1 (1-m DEM and 10-m LC) and Run 9 (30-m DEM and 1-km LC), and the greater drop-off in VDT output overlap occurs between the 12-m and 30-m DEM resolutions and is more pronounced than the change from 1- to 12-m. For specific reaches of the stream network, especially confirmed in the mainstream reach of Buncombe County, higher resolution LC consistently produced a higher percentage overlap with OPERA observed data. For CONUS applications with a wide array of higher resolution data available, users may opt for 1-m DEM and

TABLE 7 | Percent change in flood extent area for all combinations of Run IDs from higher to lower resolutions.

Run ID 1	Run ID 2	Percent change
1	2	8.39%
1	3	13.45%
1	4	55.41%
1	5	75.69%
1	6	97.56%
1	7	234.47%
1	8	244.87%
1	9	278.05%
2	3	4.67%
2	4	43.37%
2	5	62.08%
2	6	82.26%
2	7	208.57%
2	8	218.16%
2	9	248.77%
3	4	36.98%
3	5	54.85%
3	6	74.13%
3	7	194.81%
3	8	203.98%
3	9	233.22%
4	5	13.05%
4	6	27.12%
4	7	115.22%
4	8	121.92%
4	9	143.26%
5	6	12.45%
5	7	90.38%
5	8	96.30%
5	9	115.18%
6	7	69.30%
6	8	74.57%
6	9	91.36%
7	8	3.11%
7	9	13.03%
8	9	9.62%

the highest resolution LC available, but if a larger margin of error is acceptable or a single parameter like top-width is extracted for an isolated analysis, 12-m DEM may satisfy user needs at

a potentially lower computational cost. AutoRoute users can therefore be better informed on which of the input product spatial resolutions best satisfy modeling needs.

While modeling limitations significantly impact the percent overlap results, the most influential factor behind these results was the timing difference between the peak flows and Sentinel-10's data collection. Since the true peak flow extent simulated by the AutoRoute model was not captured under Sentinel-1's orbit, model performance relative to observations cannot be fully quantified. However, comparisons of percent change across combinations of DEM and LC resolutions helped quantify the variability observed between input data resolution and resulting modeled flood extents. This analysis provides insight into the potential tradeoffs of using different datasets, which is particularly valuable for informing modeling efforts in data-sparse regions where the highest resolutions may not be available.

Because neither the selection of units (raster cells within the AOI) nor the allocation of units to groups (model runs defined by DEM and LC resolution) was done using a random chance mechanism, complete causality nor population inferences can be drawn with certainty. This analysis only considers a single area of interest for one flood event and flow scenario, so statistical conclusions may not be generalizable for AutoRoute over any AOI or flow scenario. These conclusions may be confirmed in future work by repeating this methodology over additional AOIs, flood events, and flow scenarios. This study does, however, validate a pilot study with similar results conducted over a region of the South Platte River in Northeastern Colorado using low, medium, and high flow scenarios.

AutoRoute also frequently estimates flood inundation for data-sparse OCONUS areas of interest. When this is the case, users may attempt to mitigate one low spatial resolution input product with a higher spatial resolution for the other. While DEM has proven to be the more influential product in VDT output, every increase in spatial resolution for both DEM and LC results in improved overlap with OPERA observed flood data. For AutoRoute applications in data-sparse regions, users may reference this study to roughly estimate effect size differences. For an AOI similar to Asheville, North Carolina, and for a flow scenario similar to that used in this study, users may estimate approximately 70% nonoverlap between velocity outputs of an AutoRoute run with 1-m DEM and 10-m LC and a run with 30-m DEM and 1-km LC with a probability of 0.91 that the former produces higher velocity values than the latter. Creation of synthetic streamflow datasets that more closely match observed flows recorded during Hurricane Helene may provide additional insights to the relationship between various input datasets and reduce potential errors caused by the differences observed from input flows and observed flows used in this study.

This study emphasizes the importance and need of accurate and timely flood extent observational tools for advancing flood-model evaluation, especially for historic events. Alternative observational tools such as high-water marks, aerial imagery, and land disturbance rasters may provide improved comparisons with AutoRoute flood extents; however, their accessibility and availability vary greatly and comparisons with such are beyond

the scope of this paper. More accurate observational datasets will help to better inform model accuracy assessments and aid in identifying additional improvements in forecast-based systems and retrospective analyses. Increased understanding of existing models and frameworks will support model improvements, enhancing performance and providing decision makers with more timely and accurate products. Improved flood-inundation modeling accuracy will ultimately inform increasingly critical planning and resilience efforts due to the increased intensity and severity of storms and resulting impacts (Knutson et al. 2021).

Author Contributions

Joshua Smallwood: formal analysis, methodology, data curation, writing – original draft, review and editing. Robert Pennell: formal analysis, methodology, writing – original draft, review and editing. Emily S. Ondich: conceptualization, funding acquisition, supervision, writing – review and editing.

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Disclosure

DEM resolution is more influential than land cover on first-order flood inundation modeling results. Accurate and timely flood observation data, especially for peak events, remain a crucial need.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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